

Improving Production Planning through Material Flow Analysis: Gas Stove Parts Manufacturer

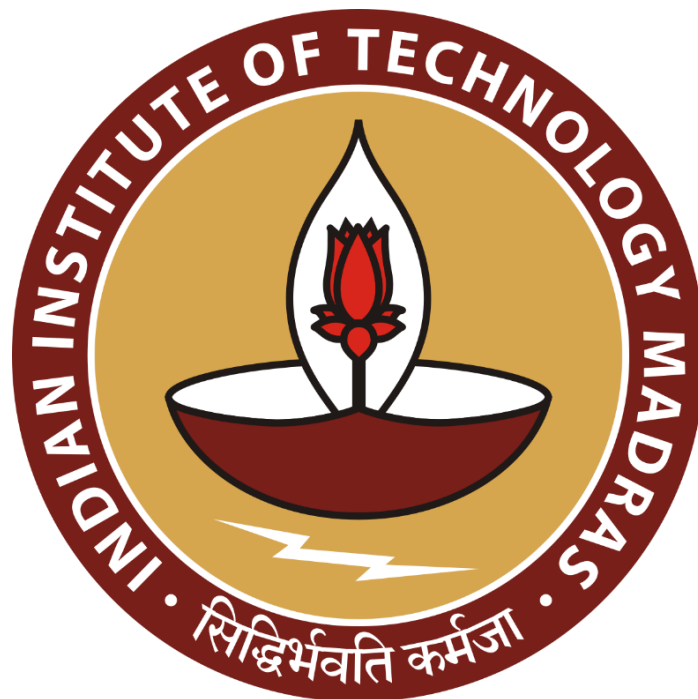
A Mid-Term report for the BDM capstone Project

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1. Executive Summary

Improving Production Planning through Material Flow Analysis: A Case Study of Metafab India

Metafab India is an MSME-based B2B manufacturing enterprise located in Greater Noida, Uttar Pradesh, producing precision gas stove components, including brass burner rings, cast iron pan supports, and gas valve assemblies. The organization maintains operational records manually through factory registers, making it difficult to integrate procurement, production, inventory, and sales information for effective decision-making. Consequently, production planning is largely experience-driven, limiting the organization's ability to evaluate material utilization, identify operational bottlenecks, align production with customer demand, and optimize inventory. This project aims to develop a quantitative, data-driven production planning framework to improve operational efficiency and managerial decision-making.

The study is based entirely on **primary data** collected from the organization and manually digitized into a structured master database. The database comprises **eight operational datasets** with over **2,500 observations** collected across approximately six months, covering procurement, production, sales, raw material inventory, finished goods inventory, worker attendance, machine breakdowns, and production capacity. The metadata includes transaction dates, material and product details, quantities, production output, inventory movements, labour availability, machine downtime, and sales values. Descriptive statistical analysis was performed to summarize operational characteristics and validate data quality.

The analytical framework integrates **Material Flow Analysis (MFA)**, **descriptive analytics**, **trend analysis**, **comparative analysis**, and **operational performance evaluation** to examine relationships among procurement, production, inventory, and customer demand. Preliminary findings reveal noticeable variations in material flow, production output, inventory movement, and sales patterns, highlighting opportunities to improve production planning and resource utilization. The final phase will focus on developing practical, data-driven recommendations for Metafab India's production planning process.

2. Proof of Data Originality

The present study is based entirely on **primary data** collected directly from **Metafab India**, a B2B manufacturer of gas stove components located in Greater Noida, Uttar Pradesh. All operational records were obtained with the organization's permission and manually digitized from handwritten factory registers into a structured Microsoft Excel master dataset. No secondary datasets or publicly available databases have been used in this project.

The originality of the collected data is supported through the following evidence:

S. No.	Evidence Submitted	Remarks / Description	Google Drive Link
1	Organization Authorization Letter	Official permission letter from Metafab India on company letterhead with stamp and authorized	(Google Drive Link)

S. No.	Evidence Submitted	Remarks / Description	Google Drive Link
		signature permitting data collection for the BDM project.	
2	Organization Registration / License	Manufacturing registration/license establishing the legal identity and authenticity of the organization.	(Google Drive Link)
3	Master Dataset	Digitized Microsoft Excel master dataset created by manually converting handwritten operational records into structured datasets for analysis.	(Google Drive Link)
4	Factory Images	Photographs of the manufacturing facility, production area, machinery, inventory, and operational activities (maximum five images).	(Google Drive Link)
5	Interaction Video	Video (3–8 minutes) demonstrating interaction with the organization's point of contact discussing the business operations and project.	(Google Drive Link)
6	Live Operations Videos	Video recordings capturing live manufacturing operations and workflow within the factory premises.	(Google Drive Link)
7	Interview Transcript	Transcript of the discussion conducted with the business owner regarding production planning, manufacturing process, and operational challenges.	(Google Drive Link)
8	Complete Evidence Repository	Consolidated Google Drive folder containing all supporting documents, datasets, images, videos, and project evidence.	(Google Drive Link)

3. Metadata and Descriptive Statistics

3.1 Metadata Summary, Variable Justification and Relevance to the Business Problem

The study utilized eight primary operational datasets collected directly from Metafab India. The selected variables collectively represent procurement, production, inventory, sales, workforce, machine performance, and production capacity. Each dataset was selected because it contributes directly to evaluating production planning, material flow, inventory utilization, operational efficiency, and customer demand.

Table 3.1: Metadata Summary, Variable Justification and Business Relevance

Dataset	Key Variables Collected	Business Purpose	Relevance to Production Planning / Problem Statement
Purchase Data	Purchase Date, Material Name, Quantity Purchased, Unit Price, Total Cost	Evaluate procurement activities	Assesses procurement planning, purchasing decisions, and raw material availability for production.
Production Data	Production Date, Product Name, Raw Material Consumed, Good Units Produced, Defective Units	Measure manufacturing performance	Evaluates production efficiency, material utilization, production variability, and quality performance.
Sales Invoice Data	Invoice Date, Customer, Product, Quantity Sold, Selling Price, Total Amount	Analyse customer demand	Enables comparison of production output with customer demand and order fulfilment.
Raw Material Inventory	Opening Stock, Purchased Quantity, Consumed Quantity, Closing Stock	Monitor inventory movement	Supports Material Flow Analysis by evaluating stock availability, inventory utilization, and material movement.
Finished Goods Inventory	Opening Stock, Production, Sales, Closing Stock	Track finished goods	Assesses inventory accumulation and alignment between production and market demand.
Workers Attendance Log	Workers Planned, Workers Present, Overtime Hours	Assess labour availability	Identifies workforce constraints affecting production continuity and scheduling efficiency.
Machine Breakdown Log	Machine, Breakdown Reason, Downtime, Production Affected	Analyse equipment performance	Identifies operational bottlenecks and production interruptions caused by machine failures.
Production Capacity	Product, Maximum Daily Capacity, Practical Capacity, Machine Used	Evaluate manufacturing capability	Compares available production capacity with actual production to assess capacity utilization and production planning effectiveness.

Overall Justification

The selected metadata directly supports the project's objective of improving production planning through Material Flow Analysis. Procurement, production, inventory, and sales

variables enable analysis of material movement across the manufacturing process, while workforce, machine, and capacity data help identify operational constraints affecting production continuity. Together, these datasets provide the quantitative foundation for evaluating production planning efficiency and developing data-driven operational recommendations.

3.3 Summary of Descriptive Statistics along with Justification

Dataset	Key Statistics	Business Justification
Purchase Data	100 purchase transactions; Average quantity = 967.25 kg ; Average procurement cost = ₹149,413.59 ; Standard deviation = 707.55 kg	High variability in procurement quantities indicates demand-driven purchasing, making procurement planning critical for inventory optimization.
Production Data	Total production = 185,173 units ; Average daily production = 399.08 units ; Average raw material consumption = 66.43 kg ; Average defects = 19.38 units	Production statistics establish the manufacturing baseline and help evaluate production consistency, material utilization, and process efficiency.
Sales Data	315 customer invoices; Average quantity sold = 221.82 units ; Average invoice value = ₹15,337.86 ; Standard deviation = 177.59 units	Variations in customer order size indicate fluctuating demand, emphasizing the need to align production with sales requirements.
Raw Material & Finished Goods Inventory	Average raw material inventory = 6,435 kg ; Average finished goods inventory = 1,071 units	Continuous inventory movement reflects the interaction between procurement, production, and sales, providing the foundation for Material Flow Analysis (MFA).
Workforce & Machine Performance	Average workers present = 13.88 (planned = 15.73); Average overtime = 2.46 hours/day ; Average machine downtime = 4.06 hours/breakdown	Labour availability and machine reliability directly influence production continuity, scheduling efficiency, and overall operational performance.

Overall Statistical Observations

The descriptive statistical analysis reveals several important operational characteristics:

- Procurement quantities and procurement costs exhibit considerable variability, indicating flexible purchasing decisions aligned with production requirements.

- Production output remains relatively stable despite fluctuations in procurement, suggesting that inventory plays an important role in maintaining manufacturing continuity.
 - Customer demand varies across transactions, resulting in differences in sales quantities and revenue over the observation period.
 - Continuous movement in raw material and finished goods inventories reflects the interconnected nature of procurement, production, and sales operations.
 - Workforce availability, overtime, and machine downtime introduce additional operational variability that may affect production planning and manufacturing efficiency.
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4. Explanation of Analysis Process

4.1 Data Cleaning and Preprocessing

Since Metafab India maintains operational records in handwritten registers, the primary data was first digitized into Microsoft Excel and subsequently cleaned and standardized to ensure accuracy and consistency.

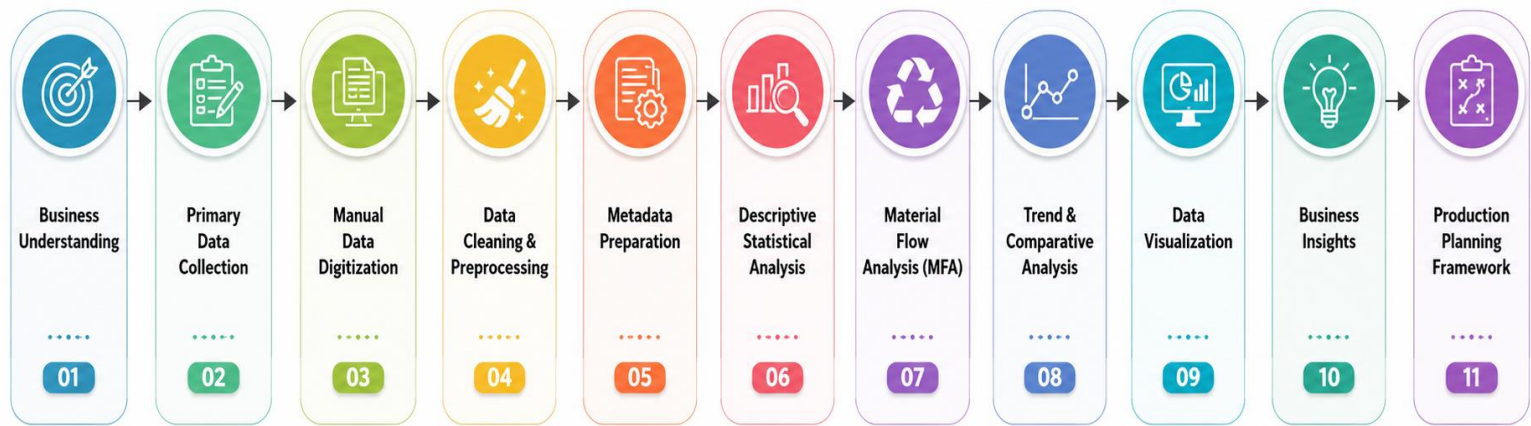
The following preprocessing activities were performed:

1. **Manual Digitization:** Procurement, production, sales, inventory, attendance, and machine breakdown records were converted into structured spreadsheets.
2. **Data Validation:** All entries were cross-verified with the original factory registers to eliminate transcription errors.
3. **Handling Missing Values:** Missing observations were validated against source records and retained as missing where original data was unavailable.
4. **Duplicate Verification:** Duplicate records were identified and removed to prevent double counting.
5. **Standardization:** Material names, product names, units of measurement, and date formats were standardized across all datasets.
6. **Data Type Conversion:** Quantitative variables were converted into numeric format, while dates were formatted uniformly.
7. **Feature Preparation:** Additional variables, including month and total transaction value, were created to support trend analysis and business KPI computation.

Importance of Data Cleaning

Data cleaning improved the reliability and consistency of the operational datasets by eliminating errors, duplicate records, and formatting inconsistencies. As the analysis was based entirely on manually maintained primary data, preprocessing was essential to ensure accurate statistical analysis and meaningful business insights.

4.2 Analysis Workflow



4.3 Analysis Methods

Analysis Method	Mathematical Definition / Formula	Justification	Business Objective
Descriptive Statistics	$\bar{x} = \frac{\sum x}{n}, s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$	Summarizes central tendency and variability of operational data.	Understand procurement, production, inventory and sales behaviour.
Trend Analysis	Time-series comparison of monthly observations	Identifies temporal changes in procurement, production and sales.	Evaluate production planning consistency.
Material Flow Analysis (MFA)	Input = Output + Accumulation + Loss	Evaluates movement of materials through the manufacturing process.	Assess synchronization between procurement, production and sales.
Comparative Analysis	Product-wise comparison of actual output with production capacity	Identifies production gaps and resource utilization.	Evaluate production efficiency.
Coefficient of Variation	$CV = \frac{\sigma}{\mu} \times 100$	Measures production variability independent of scale.	Assess production consistency.

Analysis Method	Mathematical Definition / Formula	Justification	Business Objective
Operational Bottleneck Analysis	Comparison of downtime, defects and labour availability	Identifies constraints affecting manufacturing operations.	Improve production continuity.
Root Cause Analysis (RCA)	Ranking operational issues based on performance indicators	Identifies underlying causes of operational inefficiencies.	Support data-driven managerial decisions.
Dashboard-Based KPI Analysis	Integration of operational KPIs	Provides a consolidated performance overview.	Support strategic production planning.

4.4 Analysis Procedure

The analytical process was executed in the following sequence:

1. Primary operational records were collected directly from Metafab India.
 2. The handwritten records were digitized into structured datasets.
 3. Data cleaning and preprocessing were performed to ensure quality and consistency.
 4. Metadata was prepared to define variables and establish their relationship with the business problem.
 5. Descriptive statistical measures were computed to summarize operational performance.
 6. Procurement and production datasets were analysed to evaluate purchasing and manufacturing patterns.
 7. Material Flow Analysis was conducted to examine the movement of materials across procurement, production, and sales.
 8. Trend and comparative analyses were performed to identify operational variations and production capacity utilization.
 9. Machine downtime, labour attendance, and production defects were analysed to identify operational bottlenecks.
 10. Root Cause Analysis and KPI-based dashboard analysis were conducted to generate preliminary business insights for improving production planning.
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5. Results and Findings

This section presents the preliminary findings derived from the analytical framework discussed in Section 4. The findings integrate evidence from Material Flow Analysis (MFA), descriptive statistics, trend analysis, comparative analysis, and operational performance evaluation to identify operational patterns influencing production planning.

5.1 How effectively are procurement and production synchronized?

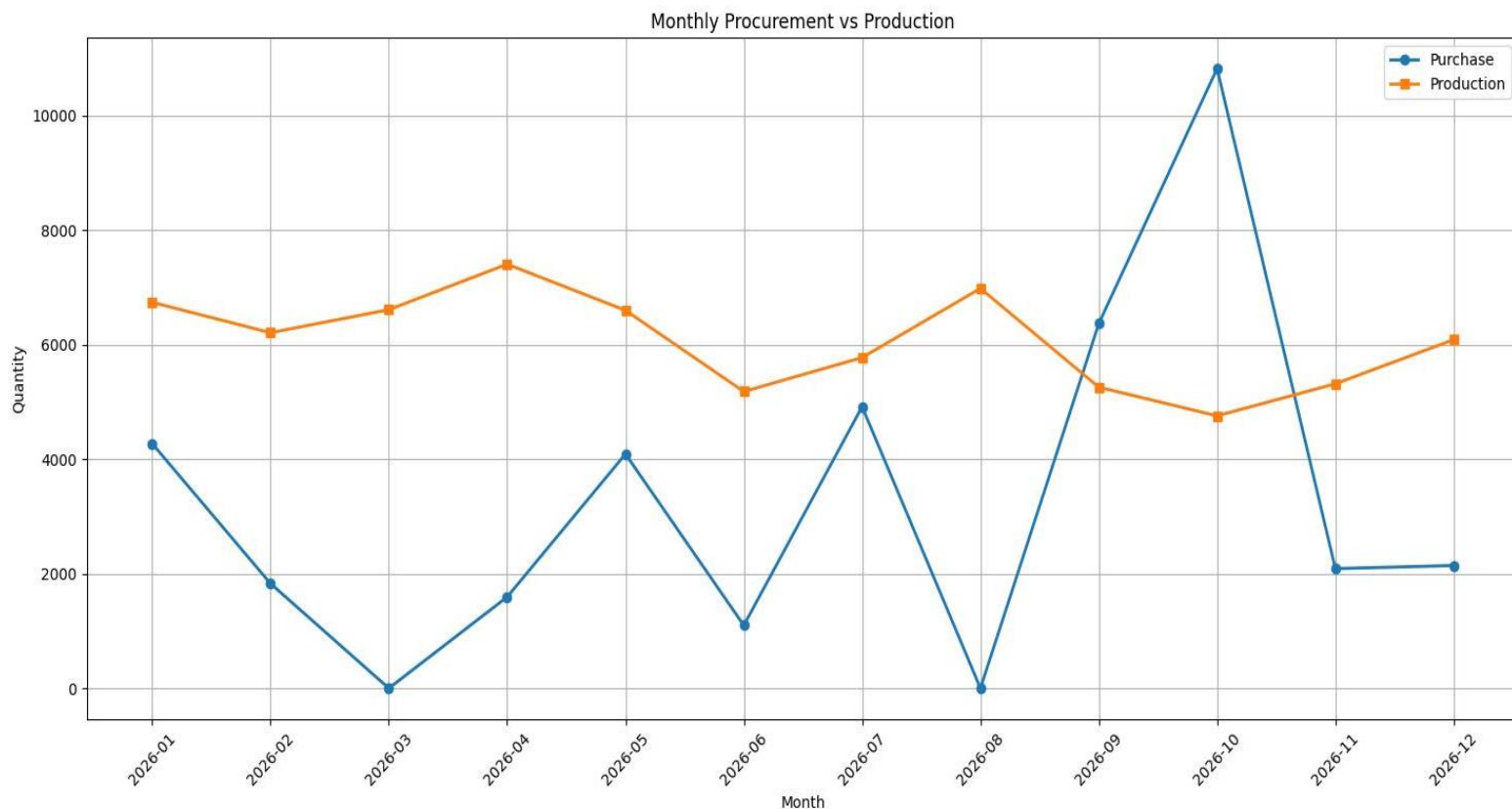


Figure 5.1 Monthly Procurement vs Production Trend

Trends and Patterns

- Procurement fluctuated considerably throughout the six-month period, whereas production remained comparatively stable.
- Procurement reached its highest level in April, while production exhibited only moderate variation.
- Sales remained consistently lower than production, indicating temporary accumulation of finished goods inventory.

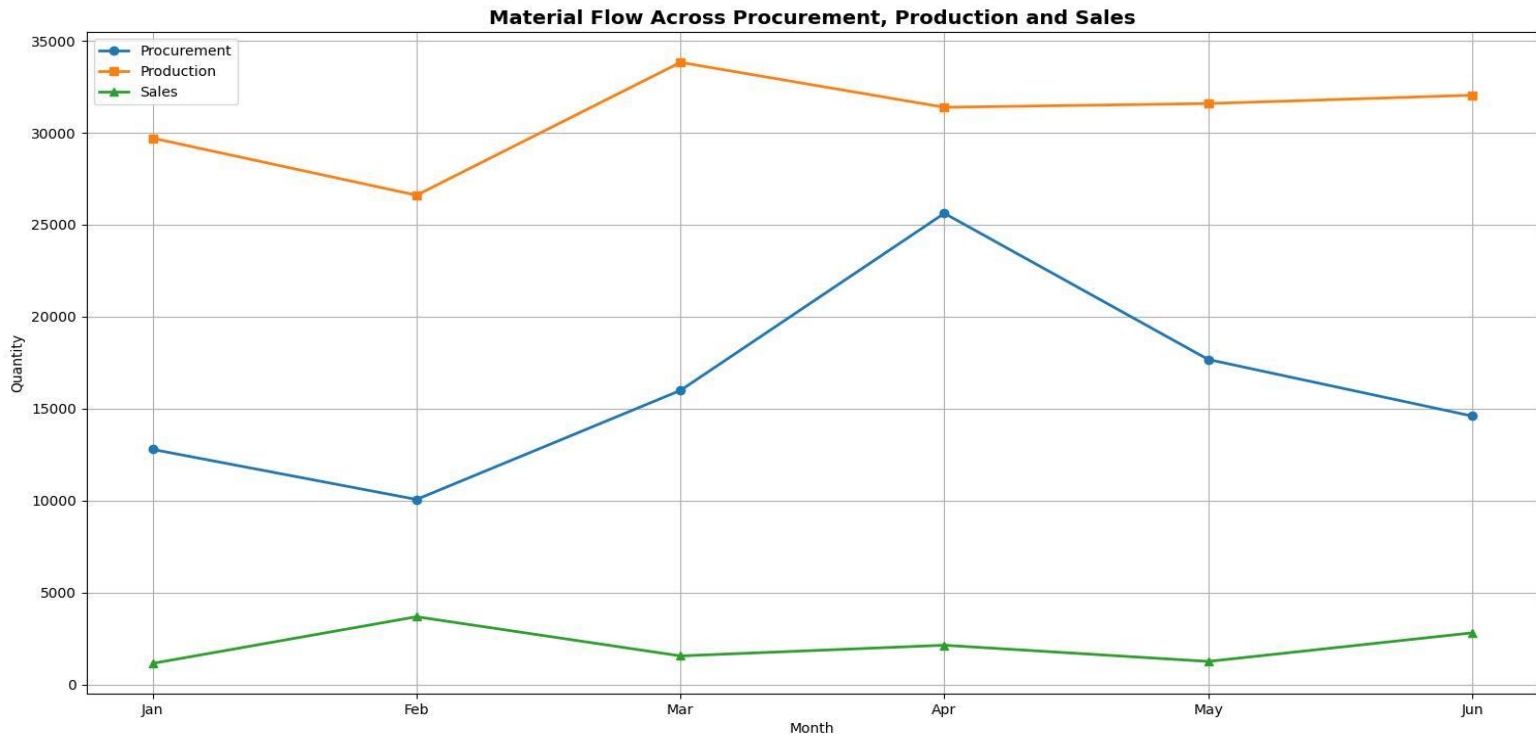


Figure 5.2 Material Flow Analysis

Preliminary Insights

The observed patterns suggest that production continuity is supported by inventory rather than immediate procurement. Material Flow Analysis indicates opportunities to improve synchronization between procurement, production scheduling, and customer demand to reduce inventory holding while maintaining uninterrupted manufacturing operations.

5.2 What factors influence production performance?

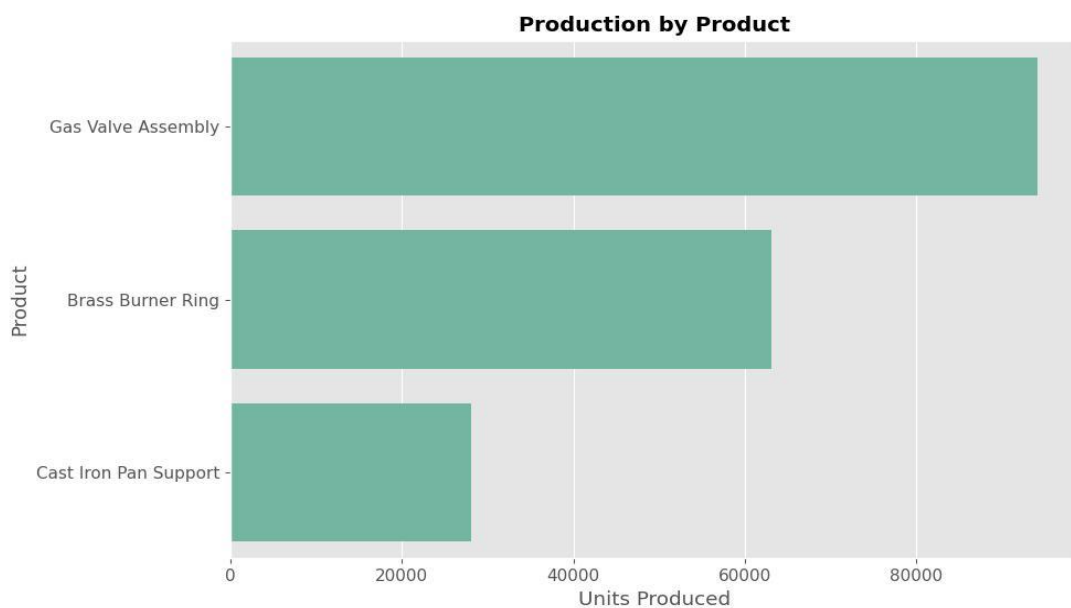


Figure 5.3 Product-wise Production Output

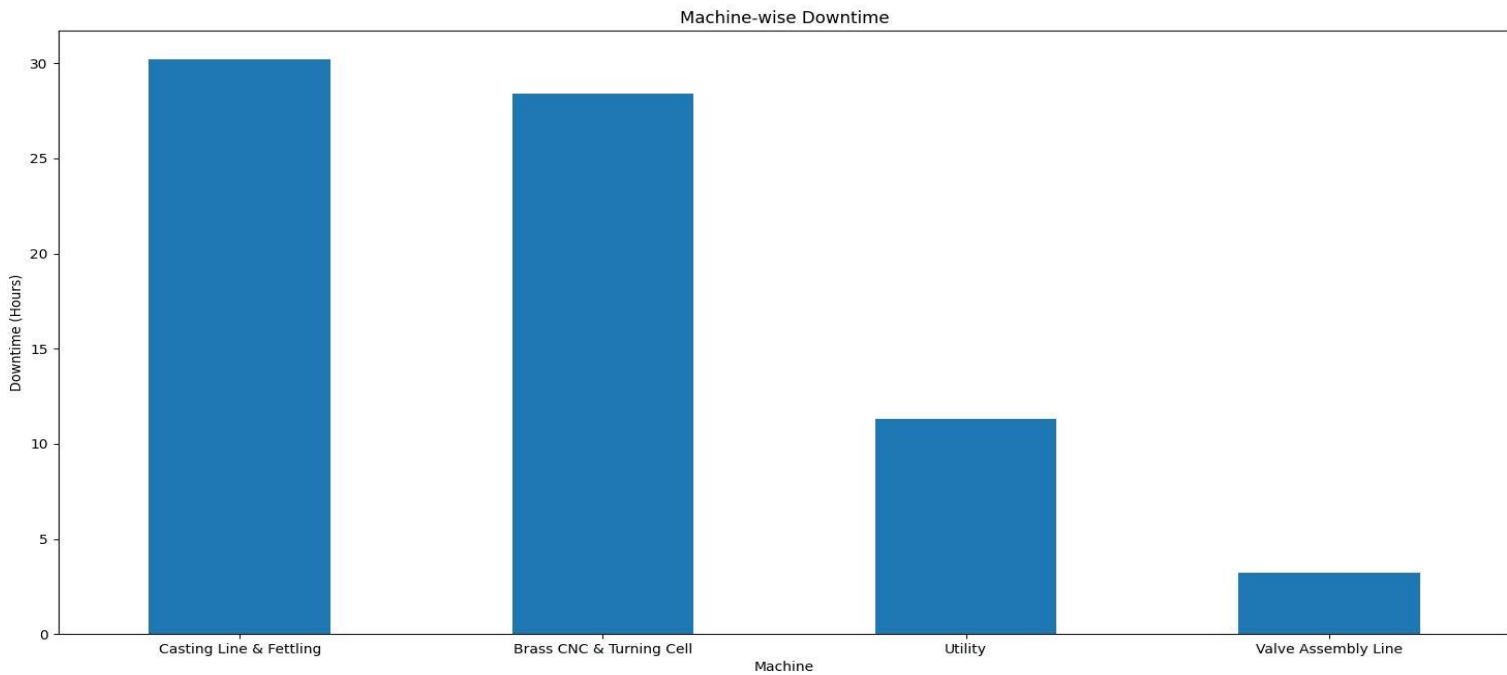


Figure 5.4 Operational Bottleneck Analysis

Trends and Patterns

- The organization produced **185,173 good units** over **180 production days**, with Gas Valve Assembly contributing the largest share of production.
- Machine breakdowns resulted in **73.10 hours** of downtime, while average worker attendance remained **88.20%**.
- The production defect rate was **4.63%**, indicating moderate material losses during manufacturing.

Preliminary Insights

Although production output remained relatively stable, workforce availability, machine reliability, and production defects collectively influence manufacturing continuity. These operational constraints should be incorporated into future production planning decisions.

5.3 How efficiently are manufacturing resources utilized?

Trends and Patterns

- Production capacity utilization varied across product categories.
- Manufacturing resources were primarily allocated to high-demand products, with Gas Valve Assembly accounting for the largest production volume.
- Actual production did not fully utilize practical production capacity across all products.

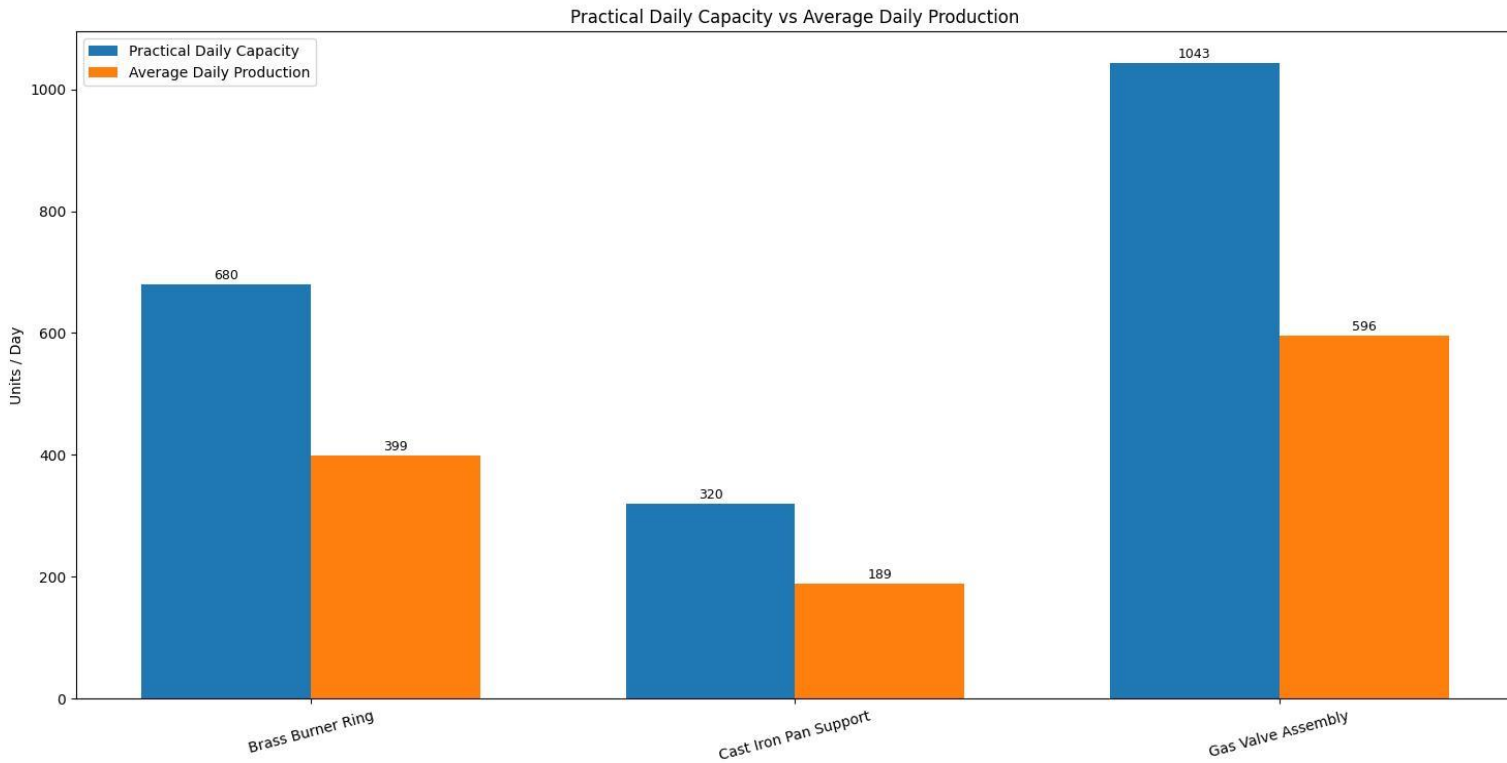


Figure 5.5 Practical Capacity vs Actual Production

Preliminary Insights

The findings indicate opportunities to improve capacity utilization by aligning production schedules with product demand and available manufacturing resources.

5.4 Overall Operational Performance

Trends and Patterns

- Total procurement expenditure exceeded **₹14.94 million**, resulting in the production of **185,173 good units** and sales revenue of **₹4.83 million**.
- Procurement, production, inventory, workforce, quality, and maintenance exhibited measurable operational variability.
- The analyses collectively demonstrate that production planning depends on the coordinated performance of multiple operational functions rather than any single business process.

Preliminary Insights

The integrated operational assessment highlights that improving production planning requires synchronized decision-making across procurement, inventory management, production scheduling, workforce planning, and preventive maintenance. These preliminary findings provide the foundation for developing data-driven recommendations in the final phase of the project.